BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 11 — Basic Statistics

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question	Answer	Key Explanation
1	Data not arranged in systematic order is called:	**B**	**1 mark** **Raw data (ungrouped data)** is data in its original, unordered form. **Grouped data** is the arranged/tabulated form. Distractor D (discrete data) refers to a data type, not arrangement status.
2	Class mark of 21–30 is:	**C**	**1 mark** **Class mark = (lower limit + upper limit) / 2 = (21 + 30)/2 = 25.5.** Common error: choosing 25 (which is the class mark of 20–30, not 21–30).
3	Lower class boundary of 31–40 is:	**C**	**1 mark** **Lower boundary = lower limit − 0.5 = 31 − 0.5 = 30.5.** The adjustment factor is ½ of the gap between the upper limit of one class and the lower limit of the next (both are 1 apart here).
4	Cumulative frequency is the sum of the class frequency and frequencies of all:	**B**	**1 mark** **Preceding classes.** We accumulate from the start; the c.f. grows as we move down the table. Taking succeeding classes (A) gives the reverse — a common confusion.
5	In a histogram with unequal class intervals, bar height represents:	**C**	**1 mark** **Frequency Density (F.D) = frequency / class interval width.** Using raw frequency (A) would make wider bars misleadingly taller. Area of each bar = frequency.
6	Arithmetic mean of 2, 4, 6, 8, 10 is:	**B**	**1 mark** **$\Sigma x = 30$, $n = 5$, $\bar{x} = 30/5 = 6$.** Distractor A (5) is the count of values, not the mean. Distractor C (7) results from adding incorrectly.
7	Arithmetic mean for grouped data uses class marks formula:	**B**	**1 mark** **$\bar{x} = \Sigma fx / \Sigma f$** where x = class mark and f = frequency. $\Sigma x/n$ (A) is for ungrouped data. Option C has the ratio inverted.
8	Median of 3, 5, 7, 9, 11 is:	**B**	**1 mark** n = 5 (odd), so median = ((5+1)/2)th = 3rd value. Arranged: 3, 5, **7**, 9, 11. Median = **7**. Distractor C (8) is the mean of 7 and 9.
9	In Mode formula, ‘l’ stands for:	**B**	**1 mark** **l = lower class boundary of the modal class.** Not the class limit (D) — boundaries differ from limits by the adjustment factor 0.5. Class mark (A) is the mid-value, which is different.
10	Mode = 26, mean = 23; find median:	**B**	**1 mark** Empirical relation: median = ⅓(mode + 2 mean) = ⅓(26 + 46) = ⅓(72) = **24**. Distractor A uses mode − mean incorrectly.
11	Weighted mean formula is:	**B**	**1 mark** **$\bar{x}_w = \Sigma wx / \Sigma w$.** Weights replace frequencies. Option A inverts the ratio. Options C and D are structurally wrong.
12	In a frequency polygon, points are plotted against:	**C**	**1 mark** **Class marks (mid-values)** on the horizontal axis, with **frequencies** on the vertical axis. Class boundaries (A) are used for histograms. Distractor D (cumulative frequency) describes an ogive.

SECTION B — Short Question Answers (11 × 3 = 33 Marks)

2(i) — Library Data / Class Terminology 3 marks

(a) Total readers = 5 + 4 + 8 + 9 + 2 + 2 = **30** 1

(b) Group with highest frequency: **31–40** (f = 9) 1

(c) Lower boundary of last class (51–60): lower limit − 0.5 = 51 − 0.5 = **50.5** 1

OR

(a) Raw data: Data not arranged in any systematic order. 1

(b) Class limits: The two boundary numbers defining a class (lower class limit = smaller, upper class limit = larger). For 1–10: lower = 1, upper = 10. 1

(c) Class boundaries: Formed by subtracting/adding the adjustment factor (0.5 here) to class limits. For 1–10: boundaries are 0.5–10.5. Adjustment factor = ½(11−10) = **0.5**. 1

2(ii) — Class Interval 1-9 / Frequency Density 3 marks

(a) Lower class limit = **1**

(b) Upper class limit = **9**

(c) Adjustment factor = ½(10 − 9) = 0.5. Boundaries: lower = 1 − 0.5 = **0.5**, upper = 9 + 0.5 = **9.5** → class boundaries: **0.5–9.5** 1

(d) Class mark = (1 + 9)/2 = **5** 1

(Note: 1 mark for both limits + boundaries together, 1 mark for class mark.)

OR

Frequency Density (F.D): When class intervals are unequal, heights of histogram bars are not equal to frequency. To keep area proportional to frequency:
F.D = frequency / width of class interval 1

It is also called the ratio frequency/width. The y-axis of such a histogram shows F.D, and x-axis shows class boundaries. 1

Calculation: f = 12, width = 4 → F.D = 12/4 = **3** 1

2(iii) — Discrete Frequency Distribution (Die) 3 marks

No. of dots	Frequency (f)	Cumulative Frequency (c.f.)
1	4	4
2	6	10
3	5	15
4	5	20
5	4	24
6	6	30
Total	**30**	

1 mark: correct frequency column; 1 mark: correct c.f. column; 1 mark: total = 30. 3

OR

Tally Bar Method (6 steps): 3 marks

Step I: Decide the number of groups (usually 5–15).

Step II: Find approximate class interval size: $h \approx (x_l - x_s) / n$.

Step III: Draw table; write class limits in first column according to size h.

Step IV: Go through ungrouped data one by one; draw tally marks in the second column. Every 5th mark is drawn diagonally across the previous four.

Step V: Count tally marks; write frequency in the third column.

Step VI: Sum all frequencies; verify Σf = total number of values (n).

2(iv) — Arithmetic Mean (Ungrouped) **3 marks**

$x = 2, 4, 6, 8, 10, 12;\quad n = 6$

$\Sigma x = 2 + 4 + 6 + 8 + 10 + 12 = \mathbf{42}$ **1**

$\bar{x} = \Sigma x / n = 42 / 6 = \mathbf{7}$ **2**

OR

$y = 3, 4, -1, 7, -8, -5, 0;\quad n = 7$

$\Sigma y = 3 + 4 + (-1) + 7 + (-8) + (-5) + 0 = \mathbf{0}$ **1**

$\bar{y} = 0 / 7 = \mathbf{0}$ **2**

2(v) — Mean of Discrete Freq. Distribution / Grouped Mean (Fuel) **3 marks**

x	f	fx
1	3	3
2	7	14
3	5	15
4	2	8
5	3	15
Total	**Σf = 20**	**Σfx = 55**

$\bar{x} = \Sigma fx / \Sigma f = 55 / 20 = \mathbf{2.75}$ **3**

OR

Class	Class mark (x)	f	fx
0–5	2.5	5	12.5
5–10	7.5	7	52.5
10–15	12.5	10	125
15–20	17.5	6	105
20–25	22.5	2	45
Total		**30**	**340**

$\bar{x} = 340 / 30 = \mathbf{11.33}$ (approx.) **3**

2(vi) — Median (Ungrouped) **3 marks**

Data: 40, 35, 45, 60, 50, 35, 40, 45, 35, 50

Arrange in ascending order: 35, 35, 35, 40, 40, 45, 45, 50, 50, 60 **1**

n = 10 (even) → Median = ½(5th value + 6th value) = ½(40 + 45) = ½(85) = **42.5** **2**

OR

Prices: 1075, 1050, 2050, 1030, 2045, 1090, 1530

Ascending order: 1030, 1050, 1075, 1090, 1530, 2045, 2050 **1**

n = 7 (odd) → Median = ((7+1)/2)th = 4th value = **Rs 1090** **2**

2(vii) — Mode (Ungrouped) **3 marks**

(a) 30, 25, 40, 30, 45, 25, 30, 40, 30: 30 appears 4 times (most frequent). Mode = **30**. Data is **unimodal**. **1**

(b) 1, 2, 3, 2, 1, 3, 1, 2: 1 appears 3 times, 2 appears 3 times, 3 appears 2 times. Modes = **1 and 2**. Data is **bimodal**. **2**

OR

125, 130, 115, 125, 135, 120, 135, 130, 120

120 appears 2 times; 125 appears 2 times; 130 appears 2 times; 135 appears 2 times. All four occur equally. Modes = **120, 125, 130, 135**. Data is **quadrimodal / multimodal**. **3**

Note: textbook states 120, 125, 130 are three modes (trimodal); either answer accepted if reasoning shown.

2(viii) — Empirical Relation 3 marks

Given: mean = 29, median = 32

Mode = 3 Median − 2 Mean 1

Mode = 3(32) − 2(29) = 96 − 58 = **38** 2

OR

Given: mode = 26, mean = 23

Median = ⅓(mode + 2 mean) 1

Median = ⅓(26 + 2 × 23) = ⅓(26 + 46) = ⅓(72) = **24** 2

2(ix) — Weighted Mean 3 marks

Subject	x (Marks)	w (Weight)	wx
Maths	90	4	360
Physics	75	3	225
Chemistry	60	2	120
Biology	50	1	50
Total		**Σw = 10**	**Σwx = 755**

$\bar{x}_w$ = Σwx / Σw = 755 / 10 = **75.5** 3

OR

Weighted mean: Arithmetic mean where values have different levels of importance (weights) instead of equal frequencies. Formula: $\bar{x}_w$ = Σwx / Σw. 1

Item	x (Price)	w	wx
Item 1	200	5	1000
Item 2	350	3	1050
Item 3	500	2	1000
Total		**10**	**3050**

$\bar{x}_w$ = 3050 / 10 = **Rs 305** 2

2(x) — Median and Mode from Discrete Freq. Distribution 3 marks

x	f	c.f.
1	3	3
2	4	7
3	5	12
4	2	14
5	2	16
Total	**16**	

n = 16; median class contains the (n/2)th = 8th observation. 1

From c.f. table: 8th observation lies where c.f. = 12, so **Median = 3**. 1

Highest frequency = 5 (for x = 3), so **Mode = 3**. 1

OR

x	f	c.f.
2	2	2
4	5	7
6	7	14
8	4	18
10	2	20
Total	**20**	

n = 20; 10th observation: c.f. shows 10th lies in group x = 6 (c.f. = 14). **Median = 6**. 2

Highest frequency = 7 at x = 6. **Mode = 6**. 1

2(xi) — Frequency Polygon on Histogram / Class Marks 3 marks

Frequency polygon on a histogram: 3 marks

1. Find the **mid-points (class marks)** of the top edges of all histogram bars.
2. Join these mid-points with straight line segments.
3. At **both ends**, add one extra class (with zero frequency) on each side of the x-axis. Connect the polygon to these points so it touches the x-axis, forming a **closed polygon**.

Alternatively: plot class marks vs frequencies on axes and join points; then extend to the axis at both ends to close it.

OR

Ages of workers. Class mark = (lower + upper)/2. 3 marks

Class	f	Class Mark
20–24	5	22
25–29	16	27
30–34	12	32
35–39	10	37
40–44	8	42
45–49	4	47

SECTION C — Long Question Answers (4 × 5 = 20 Marks)

3(i) — Frequency Distribution Table: Masses of Boys 5 marks

Given data: 5 classes, total n = 40. **Adjustment factor = ½(136 − 135) = 0.5.** 1

Class (Masses)	f	Class Mark	Class Boundaries	Cumulative f
125–135	8	130	124.5–135.5	8
136–146	13	141	135.5–146.5	21
147–157	11	152	146.5–157.5	32
158–168	6	163	157.5–168.5	38
169–179	2	174	168.5–179.5	40
Total	**40**			

Class marks column: 1 Class boundaries column: 2 Cumulative frequency column: 1

OR

Heights of 44 students (unequal intervals): F.D = frequency / class interval width 1

Height	f	Width (h)	F.D = f/h	Class Boundaries
130–144	9	15	0.6	129.5–144.5
145–149	7	5	1.4	144.5–149.5
150–154	8	5	1.6	149.5–154.5
155–159	6	5	1.2	154.5–159.5
160–169	10	10	1.0	159.5–169.5
170–179	4	10	0.4	169.5–179.5
Total	**44**			

F.D column: 2 Boundaries column: 2

3(ii) — Arithmetic Mean of Grouped Data (Marks) 5 marks

Formula: $\bar{x} = \Sigma fx / \Sigma f$ (using class marks) 1

Marks	f	Class Mark (x)	fx
0–10	2	5	10
10–20	5	15	75
20–30	7	25	175
30–40	3	35	105
40–50	8	45	360
Total	**Σf = 25**		**Σfx = 725**

2 for correct table $\bar{x}$ = 725 / 25 = **29** 2

OR

Detergent packs: class marks for 1–9 series: x = (lower+upper)/2. 1

Class (kg)	f	Class Mark (x)	fx
1–9	10	5	50
10–18	15	14	210
19–27	18	23	414
28–36	15	32	480
37–45	12	41	492
Total	**Σf = 70**		**Σfx = 1646**

2 for table $\bar{x}$ = 1646 / 70 = **23.51 kg** (approx.) 2

3(iii) — Median of Grouped Data (Marks of 100 Students) 5 marks

n = 100; median = (n/2)th = 50th value. 1

Marks	f	Class Boundaries	c.f.
10–19	5	9.5–19.5	5
20–29	25	19.5–29.5	30
30–39	40	29.5–39.5	70
40–49	20	39.5–49.5	90
50–59	10	49.5–59.5	100

50th value lies where c.f. first reaches or exceeds 50: that is the 30–39 class (c.f. = 70). **Median group: 29.5–39.5**. 1

l = 29.5, h = 10, f = 40, n = 100, c = 30 (c.f. of preceding class). 1

Median = l + (h/f)(n/2 − c) = 29.5 + (10/40)(50 − 30) 1

= 29.5 + (1/4)(20) = 29.5 + 5 = **34.5 marks** 1

OR

Mode of Wages Distribution

Highest frequency = 20 (wages 40–44). Modal class boundaries: **39.5–44.5**. 1

l = 39.5, f_m = 20, f_1 = 8 (preceding: 35–39), f_2 = 6 (following: 45–49), h = 5. 1

Mode = l + [($f_m - f_1$) / ($2f_m - f_1 - f_2$)] × h 1

= 39.5 + [(20 − 8) / (40 − 8 − 6)] × 5 = 39.5 + [12/26] × 5 1

= 39.5 + 60/26 = 39.5 + 2.31 = **41.81 Rupees** 1

3(iv) — Weighted Arithmetic Mean (Household Expenditure) 5 marks

Formula: $\bar{x}_w$ = Σwx / Σw 1

Item	x (Rs)	w	wx
Food	8,000	20	160,000
House Rent	5,000	10	50,000
Clothing	1,000	6	6,000
Utility Bills	900	4	3,600
Education	600	2	1,200
Miscellaneous	100	1	100
Total		**Σw = 43**	**Σwx = 220,900**

3 for complete wx column and correct totals.

$\bar{x}_w$ = 220,900 / 43 = **Rs 5,137.21** 1

OR

Weighted mean for Hanzala and Zaki:

Subject	w	Hanzala (x)	wx (H)	Zaki (x)	wx (Z)
Mathematics	4	70	280	90	360
Physics	3	95	285	55	165
Chemistry	2	80	160	50	100
Biology	1	55	55	65	65
Total	**10**		**780**		**690**

Hanzala: $\bar{x}_w$ = 780 / 10 = **78** 2

Zaki: $\bar{x}_w$ = 690 / 10 = **69** 2

Hanzala scores higher on the weighted scale (78 > 69). 1